Multi-Phase Results and Cross-Phase Analysis

Phase-Specific Findings

Foundation Phase (Phase 1) I

The configured foundation phase retained 46 records after its search, filtering, and de-duplication rules. Its taxonomy includes transit spectroscopy, emission spectroscopy, phase-curve analysis, and direct imaging. These labels describe the configured retrieval design; the synthetic fixture cannot establish how widely those approaches occur in the published literature.

JWST Phase (Phase 2) I

Phase 2 retained 17 records under the JWST-oriented queries and the configured year filter ($\geq$2020). The boundary is intended to include pre-launch calibration work and post-launch observations. It is a design choice, not evidence that the phase is exhaustive or that any instrument caused a change in the field.

Molecular Detection Phase (Phase 3) I

Phase 3 retained 29 records under five configured molecule-oriented query groups: H O, CO , CH , H S, and Na/K. The query groups operationalize the review question; they do not assert that these are the most commonly detected species or that the fixture establishes a scientific trend.

Knowledge Graph Results

Knowledge Graph Results I

The optional knowledge-graph stage reports **pending** assertions from the eligible sample. The configuration-derived hypothesis table is the authoritative mapping from review question to score:

Knowledge Graph Results I

ID	Hypothesis	Scope	Evidence score
H1	JWST Atmospheric Characterization	observational	pending
H2	Molecular Diversity Detection	chemical	pending
H3	Cross-Method Consistency	methodological	pending
H4	Theoretical-Observational Agreement	theoretical	pending

Knowledge Graph Results I

The configured scores are descriptive evidence summaries, not calibrated probabilities, causal effects, or conclusions about instrument performance. The table should be read together with the claim ledger and source-tier metadata. A fixture run cannot establish whether models and observations agree in the broader literature.

Citation Network Analysis

Citation Network Analysis I

The combined corpus contains **132** citation relationships across **70** papers, with a network density of 2.73%. This is a descriptive statistic for the retained corpus; it does not establish a field-wide citation structure or community claim.

Reproducibility Assessment

Reproducibility Assessment I

Reproducibility assessment of the sampled papers revealed a mean composite reproducibility score of pending, with 0 papers successfully scored. This analysis examines the computational workflow transparency of each paper, evaluating source data availability, method documentation, experimental reproducibility, and output accessibility.